

Young Bin Kim

M.S. STUDENT • CULTURE TECHNOLOGY, KAIST

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Research Interests

Augmented Reality, Ubiquitous Virtual Reality, Spatial Computing, Everyday Objects in Mixed Reality, On-Device 3D Object Understanding and 6DoF Tracking, Context-Aware and Proactive Augmentation

Education

Korea Advanced Institute of Science and Technology (KAIST)

M.S. CULTURE TECHNOLOGY

Daejeon, South Korea

Mar. 2025 – Feb. 2027 (expected)

- Advisor: Prof. Woontack Woo, UVR Lab

Handong Global University

B.S. AI CONVERGENCE & ENTREPRENEURSHIP / B.A. INTERNATIONAL STUDIES

Pohang, South Korea

Mar. 2020 – Feb. 2025

- Cumulative GPA: 4.06 / 4.50, Major GPA: 4.18 / 4.50, *Cum Laude*
- Undergraduate research advisor: Prof. Daseong Han, Computer Graphics Lab

Publications

UNDER REVIEW

- [R1] **Young Bin Kim** et al. First-author manuscript under review at an international conference, 2027.

INTERNATIONAL CONFERENCES AND JOURNALS

- [C1] Seok-Young Kim, Dooyoung Kim, Taejun Son, **Young Bin Kim**, Woojin Cho, Woontack Woo. 2026. Mix3R: Zero-shot Interactive 3D Scene Reconstruction from a Single Image in Mixed Reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, Proc. IEEE ISMAR 2026.
- [C2] Suji Kang, Seok-Young Kim, **Young Bin Kim**, Taewook Ha, Dieter Schmalstieg, Shohei Mori, Woontack Woo. 2026. SnapPhysics: A Physics-Aware Scene Graph from a Single View for Interactive Mixed Reality Scenes. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*.

INTERNATIONAL POSTERS

- [P1] **Young Bin Kim**, Suji Kang, Minju Baeck, Seonji Kim, Woontack Woo. 2025. Architectural Elements Augmentation from 2D Artworks for Spatial Experience of AR Exhibition. *IEEE ISMAR-Adjunct*, 861–862.

DOMESTIC CONFERENCES

- [D1] **Young Bin Kim**, Jinwoo Jeon, Woontack Woo. 2025. Instance-Aware Object-Centric 3D Map Generation Framework. *Korea Computer Congress (KCC)*, KIISE.
- [D2] Yunseop Kim, **Young Bin Kim**, Jina Lee, Jiseong Lee, Jaeho Lim, Daseong Han. 2024. A Video-based Framework for Collecting Whole-body and Hand Motion Data. *HCI Korea*, 87–91.
- [D3] Jiseong Lee, **Young Bin Kim**, Yunseop Kim, Jina Lee, Jaeho Lim, Daseong Han. 2024. Finger Motion Generation for Captured Whole Body Motion. *Korea Computer Graphics Society Conference (KCGS)*, 111–112. Poster.
- [D4] **Young Bin Kim**, Jiseong Lee, Gangsan Kim, Yunseop Kim, Jina Lee, Wonsup Lee. 2024. Measurement of VR Motion Sickness and Analysis of User Behavior Using VR Game and SSQ. *HCI Korea*, 37–42.

Research Experience

UVR Lab, KAIST

GRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. WOONTACK WOO

Daejeon, South Korea

Mar. 2025 – present

- **Object understanding and tracking for mixed reality**

- Master's thesis research on integrating everyday objects into mixed reality on the fly. First-author manuscript under review. [R1]
- Implemented the full pipeline on-device on Galaxy XR (partly under the WISE UI project).
- Contributed object reconstruction and dynamic-object tracking to a real-time dynamic digital twin (DT-XR project).
- Proposed an instance-aware, object-centric 3D map generation framework. [D1]

- **MR systems and user studies** (collaborative work)

- Built the user-study setup and analyzed the study results for a single-image 3D scene reconstruction system in MR. [C1]
- Implemented the MR system and the demo for a physics-aware scene-graph reconstruction system from a single view. [C2]

- **AR exhibition system**

- Built an AR system that augments architectural elements from 2D artworks in the physical exhibition space. [P1]

Computer Graphics Lab, Handong Global University

UNDERGRADUATE RESEARCHER, ADVISOR: PROF. DASEONG HAN

Pohang, South Korea

2023 – 2024

- Built a video-based framework for collecting whole-body and hand motion data. [D2]
- Developed a finger-motion generation method for captured whole-body motion. [D3]

Research Projects

2025 – present **WISE UI: Wearable Interface for Sustainable Enhancement** UVR Lab, KAIST

2025 – 2026 **DT-XR: Development of Dynamic Digital Twin for Realistic Untact XR Collaboration** UVR Lab, KAIST

Selected Projects

Spring 2026	ASL-based Shortcut Gestures for 3D Interaction Unity, hand tracking A 3D shortcut system mapping Ctrl+C/F/Z/X to ASL finger-spelling gestures, recognized in real time from both hands.
Fall 2025	K-Culture Stress Hunters Unity, AR; team lead An AR system for implicit stress relief through three-stage embodied interaction based on Korean cultural motifs.
2023	EasyDizzy Unity, VR [D4] A first-person VR running game with a progressive-overload system for motion-sickness adaptation; basis of a first-author user study using SSQ.
2023	Legend of XR in the Metaverse of Madness Unity, VR; team lead A three-quest VR experience for virtual-production training. Excellence Award, COSS Metaverse Contest.

Awards & Honors

2024	Grand Prize (HGU President's Award) , Pohang Culture & Arts Factory Hackathon (building-scale media facade, Unreal Engine), City of Pohang / Coding Ground / Handong Global University
2023	Creativity Award , Metaverse Asset Idea Contest, Korea Electronics Technology Institute
2023	Excellence Award , Metaverse Contest, COSS, Ministry of Education

Skills

Programming	C#, Python, Kotlin, C++
Software/Tools	Unity, Unreal Engine, OpenXR, PyTorch, OpenCV
Devices	Galaxy XR, Meta Quest 2/3, HoloLens 2
Languages	Korean (native), English (fluent)